

Homework Assignment 3 (Programming Category)

Student Name: _____

Student Session: cs6220

Posting Date: Monday of week 6 (Sept 28).

Due Date: midnight on Friday of Week 7 (Oct 9), with no penalty extension to 9am (morning) Saturday of Oct 10. It is a hard deadline. Late submission will not be accepted.

Problem: Understanding LLM Finetuning for a downstream task

You are asked to finetune one of your favorite pretrained large language model (LLM). Through this assignment you will gain experience in supervised finetuning of your chosen LLM using a [TruthfulQA benchmark dataset](https://github.com/sylinrl/truthfulqa/) (<https://github.com/sylinrl/truthfulqa/>).

TruthfulQA is a benchmark to evaluate how truthful a language model is in generating answers to questions. The benchmark comprises 817 questions. Higher performance indicates that a model is better at avoiding generating false answers learned from imitating human texts.

Requirement.

The goal of this problem is to get you to learn why and how fine-tuning can further improve the inference performance of a pretrained large language model on a downstream task, of which your chosen LLM may not perform well on some part of the benchmark data (queries).

1. The first step is to choose a pretrained LLM. Then choose 100 questions uniformly randomly as your test dataset or randomly pick up to two from each of the 38 categories, and keep the rest as your finetuning training set.
2. Perform test queries against your target LLM to collect accuracy results against correct answers from the benchmark. Recording the test queries and the results in a table (e.g., excel file).

If the target LLM has 100% success rate, then you perform another random selection of your test set. Ideally, the test performance of your target LLM should not be higher than

60-70%. Otherwise, choose another pretrained LLM and also document the results of your test for the previous choice of LLMs.

3. The third step is to choose one of the two design options as your finetuning plan:

3.1) Perform both full parameter finetuning and parameter-efficient fine-tuning using LORA (low rank adaptor, see our lecture on finetuning intro part 1) and provide comparison to three scenarios: target LLM, full parameter finetuned model, and LORA-finetuned model. Hint: your target LLM model size will be the constraint for this option.

3.2) Choose parameter efficient finetuning with LORA using two different rank (r) configurations in your LORA algorithm, provide comparison to three scenarios: target LLM, LORA finetuned model with rank r_1 and LORA-finetuned model with rank r_2 .

3. Report your finetuning training time and test accuracy for each of the two LORA versions on the same target LLM and compare your test results from each LORA finetuned model with test query results directly collected from your target LLM.

Deliverable: You are asked to produce two fine-tuned models and make the comparison.

- (1) Provide URL and a brief description of each of the following packages: (i) your open source pretrained LLM model which you have collected in Step 2 above, (ii) the benchmark dataset for supervised finetuning, (iii) the finetuning algorithm open source link, and (iv) the LORA package you choose.

- (2) **Training Time and Model Size Analysis:** You are asked to provide the model size in #parameters and bytes for your target LLM, and each of the two fine-tuned models. Discuss your experience with parameter efficient LORA finetuning, especially how the rank configuration impacts on the fine-tuned model in training time, test accuracy and model sizes.

- (3) **Test Performance Comparison and Analysis:**

(3.1) You are asked to provide a comparison table on the test results over the three models: the target LLM and the two fine-tuned models. The comparison should include avg test accuracy and avg per query test time, plus the standard derivation in time and accuracy.

(3.2) you are asked to provide four sets of example test queries and results: (i) 2 example queries that are successful by all three models, (ii) 2 example queries that are failed by the target LLM but successful by both finetuned models, (iii) 2 examples that are failed by two of the three models, and (iv) 2 examples that are failed by all three models.

(3.3) You are asked to provide screen shots of the runtime training and testing, including supervised finetuning training iterations, and hyperparameters used, error threshold at the termination.

- (4) **Final Remarks:** Elaborate three main points regarding what you have learned through this hand-on experience with supervised finetuning.

HW submission Requirement.

For all of our HW assignment, you need to follow our naming convention:

*** Reading HW should start with **HW3_P_<your last name>_<first name>**.pdf (or .zip).

The grading scale is the scale of 100 points with pass (>60), pass- (<=60) or pass+ (>90).